

OpenStax Calculus Volume 3

Section 5.5 Exercises: Triple Integrals in Cylindrical and Spherical Coordinates

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In the following exercises, evaluate the triple integrals $\iiint_B f(x, y, z) dV$ over the solid B .

1. $f(x, y, z) = z$, $B = \{(x, y, z) | x^2 + y^2 \leq 9, x \geq 0, y \geq 0, 0 \leq z \leq 1\}$
[Figure: A quarter section of a cylinder with height 1 and radius 3.]
2. $f(x, y, z) = xz^2$, $B = \{(x, y, z) | x^2 + y^2 \leq 16, x \geq 0, y \leq 0, -1 \leq z \leq 1\}$
3. $f(x, y, z) = xy$, $B = \{(x, y, z) | x^2 + y^2 \leq 1, x \geq 0, y \geq 0, x \leq y, -1 \leq z \leq 1\}$
[Figure: A wedge with radius 1, height 1, and angle $\pi/4$.]
4. $f(x, y, z) = x^2 + y^2$, $B = \{(x, y, z) | x^2 + y^2 \leq 4, x \geq 0, x \leq y, 0 \leq z \leq 3\}$
5. $f(x, y, z) = e^{\sqrt{x^2 + y^2}}$, $B = \{(x, y, z) | 1 \leq x^2 + y^2 \leq 4, y \leq 0, x \leq y\sqrt{3}, 2 \leq z \leq 3\}$
6. $f(x, y, z) = \sqrt{x^2 + y^2}$, $B = \{(x, y, z) | 1 \leq x^2 + y^2 \leq 9, y \leq 0, 0 \leq z \leq 1\}$
7. (a) Let B be a cylindrical shell with inner radius a , outer radius b , and height c , where $0 < a < b$ and $c > 0$. Assume that a function F defined on B can be expressed in cylindrical coordinates as $F(x, y, z) = f(r) + h(z)$, where f and h are differentiable functions. If $\int_a^b \tilde{f}(r) dr = 0$ and $\tilde{h}(0) = 0$, where $\tilde{f}$ and $\tilde{h}$ are antiderivatives of f and h , respectively, show that
$$\iiint_B F(x, y, z) dV = 2\pi c(b\tilde{f}(b) - a\tilde{f}(a)) + \pi(b^2 - a^2)\tilde{h}(c).$$

(b) Use the previous result to show that $\iiint_B (z + \sin \sqrt{x^2 + y^2}) dx dy dz = 6\pi^2(\pi - 2)$, where B is a cylindrical shell with inner radius π , outer radius 2π , and height 2.
8. (a) Let B be a cylindrical shell with inner radius a , outer radius b , and height c , where $0 < a < b$ and $c > 0$. Assume that a function F defined on B can be expressed in cylindrical coordinates as $F(x, y, z) = f(r)g(\theta)h(z)$, where f , g , and h are differentiable functions. If $\int_a^b \tilde{f}(r) dr = 0$, where $\tilde{f}$ is an antiderivative of f , show that
$$\iiint_B F(x, y, z) dV = [b\tilde{f}(b) - a\tilde{f}(a)] [\tilde{g}(2\pi) - \tilde{g}(0)] [\tilde{h}(c) - \tilde{h}(0)],$$

where $\tilde{g}$ and $\tilde{h}$ are antiderivatives of g and h , respectively.
(b) Use the previous result to show that $\iiint_B z \sin \sqrt{x^2 + y^2} dx dy dz = -12\pi^2$, where B is a cylindrical shell with inner radius π , outer radius 2π , and height 2.

In the following exercises, the boundaries of the solid E are given in cylindrical coordinates. Let $f(r, \theta, z)$ be the corresponding function in cylindrical coordinates.

(a) Define the region in cylindrical coordinates.

(b) Convert the integral $\iiint_E g(x, y, z) dV$ to cylindrical coordinates.

9. E is inside the right circular cylinder $r = 4 \sin \theta$, above the $r\theta$ -plane, and inside the sphere $r^2 + z^2 = 16$.
10. E is inside the right circular cylinder $r = \cos \theta$, above the $r\theta$ -plane, and inside the sphere $r^2 + z^2 = 9$.
11. E is located in the first octant and is bounded by the circular paraboloid $z = 9 - 3r^2$, the cylinder $r = \sqrt{3}$, and the plane $r(\cos \theta + \sin \theta) = 20 - z$.
12. E is located in the first octant outside the circular paraboloid $z = 10 - 2r^2$ and inside the cylinder $r = \sqrt{5}$ and is bounded also by the planes $z = 20$ and $\theta = \frac{\pi}{4}$.

In the following exercises, the function g and region E are given in rectangular coordinates.

(a) Express the region E and the function g in cylindrical coordinates. Let $f(r, \theta, z)$ be the corresponding function in cylindrical coordinates.

(b) Convert the integral $\iiint_E g(x, y, z) dV$ to cylindrical coordinates and evaluate it.

13. $g(x, y, z) = \frac{1}{x+3}$, $E = \{(x, y, z) | 0 \leq x^2 + y^2 \leq 9, x \geq 0, y \geq 0, 0 \leq z \leq x + 3\}$
14. $g(x, y, z) = x^2 + y^2$, $E = \{(x, y, z) | 0 \leq x^2 + y^2 \leq 4, y \geq 0, 0 \leq z \leq 3 - x\}$
15. $g(x, y, z) = x$, $E = \{(x, y, z) | 1 \leq y^2 + z^2 \leq 9, 0 \leq x \leq 9 - y^2 - z^2\}$
16. $g(x, y, z) = y$, $E = \{(x, y, z) | 1 \leq x^2 + z^2 \leq 9, 0 \leq y \leq 9 - x^2 - z^2\}$

In the following exercises, find the volume of the solid E whose boundaries are given in rectangular coordinates.

17. E is above the xy -plane, inside the cylinder $x^2 + y^2 = 1$, and below the plane $z = 1$.
18. E is below the plane $z = 1$ and inside the paraboloid $z = x^2 + y^2$.
19. E is bounded by the circular cone $z = \sqrt{x^2 + y^2}$ and $z = 1$.
20. E is located above the xy -plane, below $z = 1$, outside the one-sheeted hyperboloid $x^2 + y^2 - z^2 = 1$, and inside the cylinder $x^2 + y^2 = 2$.
21. E is located inside the cylinder $x^2 + y^2 = 1$ and between the circular paraboloids $z = 1 - x^2 - y^2$ and $z = x^2 + y^2$.
22. E is located inside the sphere $x^2 + y^2 + z^2 = 1$, above the xy -plane, and inside the circular cone $z = \sqrt{x^2 + y^2}$.
23. E is located inside the circular cone $x^2 + y^2 = (z - 1)^2$ and between the planes $z = 0$ and $z = 2$.
24. E is located inside the cylinder $x^2 + y^2 = 1$, above the circular cone $z = 1 - \sqrt{x^2 + y^2}$, and below the circular paraboloid $z = 1 + x^2 + y^2$, and between the planes $z = 0$ and $z = 2$.

25. [T] Use a computer algebra system (CAS) to graph the solid whose volume is given by the iterated integral in cylindrical coordinates $\int_{-\pi/2}^{\pi/2} \int_0^1 \int_{r^2}^r r \, dz \, dr \, d\theta$. Find the volume V of the solid. Round your answer to four decimal places.
26. [T] Use a CAS to graph the solid whose volume is given by the iterated integral in cylindrical coordinates $\int_0^{\pi/2} \int_0^1 \int_{r^4}^r r \, dz \, dr \, d\theta$. Find the volume V of the solid. Round your answer to four decimal places.
27. Convert the integral $\int_0^1 \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} \int_{x^2+y^2}^{\sqrt{x^2+y^2}} xz \, dz \, dx \, dy$ into an integral in cylindrical coordinates.
28. Convert the integral $\int_0^2 \int_0^y \int_0^1 (xy + z) \, dz \, dx \, dy$ into an integral in cylindrical coordinates.

In the following exercises, evaluate the triple integral $\iiint_B f(x, y, z) \, dV$ over the solid B .

29. $f(x, y, z) = 1$, $B = \{(x, y, z) | x^2 + y^2 + z^2 \leq 90, z \geq 0\}$
[Figure: A filled-in half-sphere with radius 3 times the square root of 10.]
30. $f(x, y, z) = 1 - \sqrt{x^2 + y^2 + z^2}$, $B = \{(x, y, z) | x^2 + y^2 + z^2 \leq 9, y \geq 0, z \geq 0\}$
[Figure: A quarter section of an ovoid with height 8, width 8 and length 18.]
31. $f(x, y, z) = \sqrt{x^2 + y^2}$, B is bounded above by the half-sphere $x^2 + y^2 + z^2 = 9$ with $z \geq 0$ and below by the cone $z^2 = x^2 + y^2$.
32. $f(x, y, z) = z$, B is bounded above by the half-sphere $x^2 + y^2 + z^2 = 16$ with $z \geq 0$ and below by the cone $z^2 = x^2 + y^2$.
33. Show that if $F(\rho, \theta, \phi) = f(\rho)g(\theta)h(\phi)$ is a continuous function on the spherical box $B = \{(\rho, \theta, \phi) | a \leq \rho \leq b, \alpha \leq \theta \leq \beta, \gamma \leq \phi \leq \psi\}$, then

$$\iiint_B F \, dV = \left(\int_a^b \rho^2 f(\rho) \, d\rho \right) \left(\int_\alpha^\beta g(\theta) \, d\theta \right) \left(\int_\gamma^\psi h(\phi) \sin \phi \, d\phi \right).$$

34. (a) A function F is said to have spherical symmetry if it depends on the distance to the origin only, that is, it can be expressed in spherical coordinates as $F(x, y, z) = f(\rho)$, where $\rho = \sqrt{x^2 + y^2 + z^2}$. Show that

$$\iiint_B F(x, y, z) \, dV = 2\pi \int_a^b \rho^2 f(\rho) \, d\rho,$$
where B is the region between the upper concentric hemispheres of radii a and b centered at the origin, with $0 < a < b$ and F a spherical function defined on B .
- (b) Use the previous result to show that $\iiint_B (x^2 + y^2 + z^2) \sqrt{x^2 + y^2 + z^2} \, dV = 21\pi$, where $B = \{(x, y, z) | 1 \leq x^2 + y^2 + z^2 \leq 4, z \geq 0\}$.
35. (a) Let B be the region between the upper concentric hemispheres of radii a and b centered at the origin and situated in the first octant, where $0 < a < b$. Consider F a function defined on B whose form in spherical coordinates (ρ, θ, ϕ) is $F(x, y, z) = f(\rho) \cos \phi$. Show that if $g(a) = g(b) = 0$ and $\int_a^b h(\rho) \, d\rho = 0$, then

$$\iiint_B F(x, y, z) \, dV = \frac{\pi}{2} [ah(a) - bh(b)],$$
where g is an antiderivative of f and h is an antiderivative of g .

- (b) Use the previous result to show that $\iiint_B \frac{z \cos \sqrt{x^2+y^2+z^2}}{\sqrt{x^2+y^2+z^2}} dV = \frac{3\pi^2}{2}$, where B is the region between the upper concentric hemispheres of radii π and 2π centered at the origin and situated in the first octant.

In the following exercises, the function g and region E are given in rectangular coordinates.

- (a) Express the region E and the function g in spherical coordinates. Let $f(\rho, \theta, \phi)$ be the corresponding function in spherical coordinates.
- (b) Convert the integral $\iiint_E g(x, y, z) dV$ to spherical coordinates and evaluate it.

36. $g(x, y, z) = z$; $E = \{(x, y, z) | 0 \leq x^2 + y^2 + z^2 \leq 1, z \geq 0\}$
37. $g(x, y, z) = x + y$; $E = \{(x, y, z) | 1 \leq x^2 + y^2 + z^2 \leq 4, z \geq 0, y \geq 0\}$
38. $g(x, y, z) = 2xy$; $E = \{(x, y, z) | \sqrt{x^2 + y^2} \leq z \leq \sqrt{1 - x^2 - y^2}, x \geq 0, y \geq 0\}$
39. $g(x, y, z) = z$; $E = \{(x, y, z) | x^2 + y^2 + z^2 - 2z \leq 0, \sqrt{x^2 + y^2} \leq z\}$

In the following exercises, find the volume of the solid E whose boundaries are given in rectangular coordinates.

40. $E = \{(x, y, z) | \sqrt{x^2 + y^2} \leq z \leq \sqrt{16 - x^2 - y^2}, x \geq 0, y \geq 0\}$
41. $E = \{(x, y, z) | x^2 + y^2 + z^2 - 2z \leq 0, \sqrt{x^2 + y^2} \leq z\}$
42. Use spherical coordinates to find the volume of the solid situated inside the sphere $\rho = 1$ and outside the sphere $\rho = \cos \phi$, with $\phi \in [0, \frac{\pi}{2}]$.
43. Use spherical coordinates to find the volume of the ball $\rho \leq 3$ that is situated between the cones $\phi = \frac{\pi}{4}$ and $\phi = \frac{\pi}{3}$.
44. Convert the integral $\int_{-4}^4 \int_{-\sqrt{16-y^2}}^{\sqrt{16-y^2}} \int_{-\sqrt{16-x^2-y^2}}^{\sqrt{16-x^2-y^2}} (x^2 + y^2 + z^2) dz dx dy$ into an integral in spherical coordinates.
45. Convert the integral $\int_0^4 \int_0^{\sqrt{16-x^2}} \int_{-\sqrt{16-x^2-y^2}}^{\sqrt{16-x^2-y^2}} (x^2 + y^2 + z^2)^2 dz dy dx$ into an integral in spherical coordinates.
46. Convert the integral $\int_{-2\sqrt{2}}^{2\sqrt{2}} \int_{-\sqrt{8-x^2}}^{\sqrt{8-x^2}} \int_{\sqrt{x^2+y^2}}^{\sqrt{16-x^2-y^2}} dz dy dx$ into an integral in spherical coordinates and evaluate it.
47. [T] Use a CAS to graph the solid whose volume is given by the iterated integral in spherical coordinates $\int_{\pi/2}^{\pi} \int_{5\pi/6}^{\pi/6} \int_0^2 \rho^2 \sin \phi d\rho d\phi d\theta$. Find the volume V of the solid. Round your answer to three decimal places.
48. [T] Use a CAS to graph the solid whose volume is given by the iterated integral in spherical coordinates as $\int_0^{2\pi} \int_{\pi/4}^{3\pi/4} \int_0^1 \rho^2 \sin \phi d\rho d\phi d\theta$. Find the volume V of the solid. Round your answer to three decimal places.
49. [T] Use a CAS to evaluate the integral $\iiint_E (x^2 + y^2) dV$ where E lies above the paraboloid $z = x^2 + y^2$ and below the plane $z = 3y$.

50. [T]

- (a) Evaluate the integral $\iiint_E e^{\sqrt{x^2+y^2+z^2}} dV$, where E is bounded by the spheres $4x^2 + 4y^2 + 4z^2 = 1$ and $x^2 + y^2 + z^2 = 1$.
- (b) Use a CAS to find an approximation of the previous integral. Round your answer to two decimal places.
51. Express the volume of the solid inside the sphere $x^2 + y^2 + z^2 = 16$ and outside the cylinder $x^2 + y^2 = 4$ as triple integrals in cylindrical coordinates and spherical coordinates.
52. Express the volume of the solid inside the sphere $x^2 + y^2 + z^2 = 16$ and outside the cylinder $x^2 + y^2 = 4$ that is located in the first octant as triple integrals in cylindrical coordinates and spherical coordinates.
53. The power emitted by an antenna has a power density per unit volume given in spherical coordinates by $p(\rho, \theta, \phi) = \frac{P_0}{\rho^2} \cos^2 \theta \sin^4 \phi$, where P_0 is a constant with units in watts. The total power within a sphere B of radius r meters is defined as $P = \iiint_B p(\rho, \theta, \phi) dV$. Find the total power P within a sphere of radius 20 meters.
54. Use the preceding exercise to find the total power within a sphere B of radius 5 meters when the power density per unit volume is given by $p(\rho, \theta, \phi) = \frac{30}{\rho^2} \cos^2 \theta \sin^4 \phi$.
55. A charge cloud contained in a sphere B of radius r centimeters centered at the origin has its charge density given by $q(x, y, z) = k\sqrt{x^2 + y^2 + z^2} \frac{\mu\text{C}}{\text{cm}^3}$, where $k > 0$. The total charge contained in B is given by $Q = \iiint_B q(x, y, z) dV$. Find the total charge Q .
56. Use the preceding exercise to find the total charge cloud contained in the unit sphere if the charge density is $q(x, y, z) = 20\sqrt{x^2 + y^2 + z^2} \frac{\mu\text{C}}{\text{cm}^3}$.